AWS ECR — Important Interview Questions & Answers

1.What is Amazon ECR?
Amazon ECR is a fully managed AWS service used to store, manage, and deploy container images.
It works with services such as ECS and EKS to run containerized applications.
Simple ga gurthupettuko:
ECR ante AWS lo Docker container images ni store and manage cheyyadaniki use chese service.
Eng: ECR is a managed service used to store and manage container images in AWS.

2.What is an ECR Repository?
An ECR Repository is a place where we store container images.
We can create separate repositories for different applications or services.
Simple ga gurthupettuko:
Repository ante ECR lo container images ni store chese place.
Eng: An ECR Repository is a place used to store container images.

3.What is an Image in ECR?
An image is a packaged version of an application that contains the application code and required dependencies.
ECR stores these container images so they can be used by container services.
Simple ga gurthupettuko:
Image ante application ni run cheyyadaniki required code and dependencies unna package.
Eng: An image is a package containing application code and its required dependencies.

4.What are Image Tags in ECR?
Image Tags are names used to identify different versions of a container image.
For example, we can use tags such as latest, v1, v2, or production.
Simple ga gurthupettuko:
Image Tag ante container image version ni identify cheyyadaniki use chese name.
Eng: Image Tags are used to identify different versions of container images.

5.What is the difference between ECR Repository and Image?
A Repository is the place where container images are stored.
An Image is the actual container package stored inside the repository.
Simple ga gurthupettuko:
Repository ante images store chese place, Image ante repository lo store ayye actual package.
Eng: A Repository stores images, while an Image is the actual container package.

6.What is an ECR Lifecycle Policy?
A Lifecycle Policy automatically manages old or unused container images in an ECR repository.
It can delete images based on rules such as image age or number of images.
Simple ga gurthupettuko:
Lifecycle Policy ante old or unused images ni automatic ga manage or delete cheyyadaniki use chestham.
Eng: A Lifecycle Policy automatically manages old or unused images.

7.Why do we use ECR Lifecycle Policies?
We use Lifecycle Policies to remove old images and avoid keeping unnecessary images in the repository.
This helps reduce storage usage and keeps the repository clean.
Simple ga gurthupettuko:
Old images ni remove chesi storage and repository ni clean ga maintain cheyyadaniki use chestham.
Eng: Lifecycle Policies help remove unnecessary images and manage storage.

8.What is an ECR Repository Policy?
An ECR Repository Policy controls which AWS accounts or principals can access a specific ECR repository.
It can be used to allow actions such as pulling or pushing images.
Simple ga gurthupettuko:
Repository Policy ante ECR repository ni evaru access cheyyalo control chese policy.
Eng: A Repository Policy controls access to an ECR repository.

9.What is the difference between Image Tag and Image?
An Image is the actual container package, while an Image Tag is a name used to identify a particular version of that image.
For example, an image can have a tag such as v1 or latest.
Simple ga gurthupettuko:
Image actual package, Tag aa image version ni identify chese name.
Eng: An Image is the package, while a Tag identifies its version.

10.How do you push a Docker image to ECR?
First, I would create an ECR repository and authenticate Docker with ECR.
Then I would tag the Docker image with the ECR repository URL and push the image to the repository.
Simple ga gurthupettuko:
ECR Repository create chesi, Docker login chesi, image tag chesi ECR ki push chestham.
Eng: I would create a repository, authenticate Docker, tag the image, and push it to ECR.

11.How does ECS use ECR?
ECS can pull container images from an ECR repository and use them to run ECS tasks.
The ECR image is specified in the ECS Task Definition.
Simple ga gurthupettuko:
ECS Task Definition lo ECR image specify chesi container ni run chestham.
Eng: ECS can pull ECR images and use them to run containers.

12.What happens if an ECR image is deleted?
If an image is deleted from ECR, services cannot pull that specific image anymore.
Running containers may continue running, but a new task that needs the deleted image may fail to start.
Simple ga gurthupettuko:
ECR image delete ayite new container aa image ni pull chesi start cheyyaledu.
Eng: A deleted ECR image cannot be pulled by new containers.

13.How would you secure an ECR Repository?
I would use IAM permissions and ECR Repository Policies to control access.
I would also enable image scanning and use encryption to improve container image security.
Simple ga gurthupettuko:
ECR secure cheyyali ante IAM + Repository Policy + Image Scanning + Encryption use chestham.
Eng: I would secure ECR using access control, image scanning, and encryption.

14.How would you troubleshoot an ECS task that cannot pull an ECR image?
I would check the image name and tag, ECR repository permissions, IAM role permissions, network connectivity, and whether the image exists.
Simple ga gurthupettuko:
ECR image pull problem vaste Image/Tag → IAM → Repository Policy → Network → Image existence check chestham.
Eng: I would check the image, permissions, IAM role, network, and repository.

15.What is the most important thing to remember about ECR?
ECR is used to securely store and manage container images in AWS.
The important concepts are Repositories, Images, Image Tags, Lifecycle Policies, and Repository Policies.
Simple ga gurthupettuko:
ECR ante AWS lo container images ni store, manage and secure cheyyadaniki use chese service.
Eng: ECR is used to store, manage, and secure container images in AWS.